

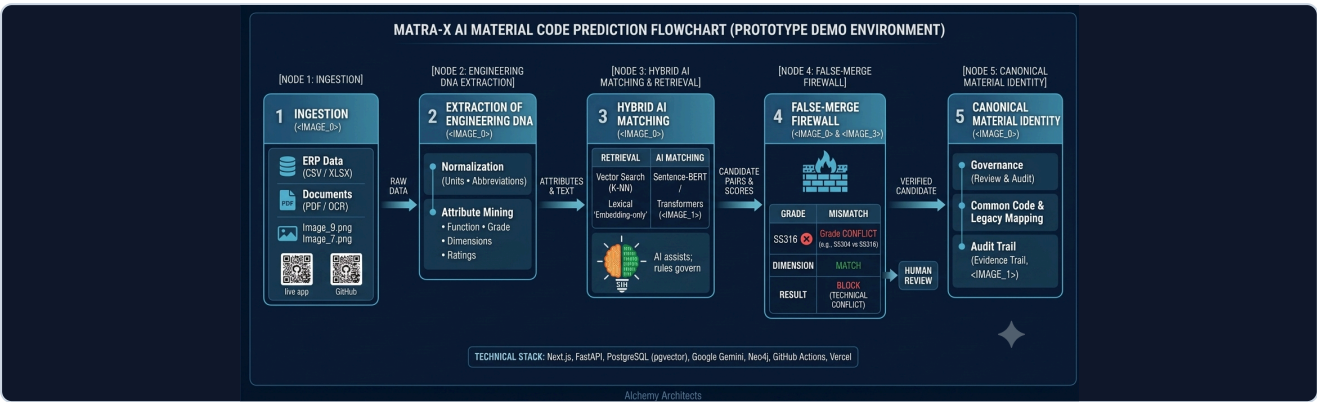
MATRA-X: Technical Prediction Report

PS ID: SIH26099 Team: Alchemy Architects **CPSE Standardization**

Core Methodology: MATRA-X harmonizes material codes across Central Public Sector Enterprises (CPSEs) using a **Hybrid AI + Structured Engineering Control** architecture. The core principle establishes that *semantic similarity is evidence, not proof*—critical engineering attribute mismatches strictly override text match scores.

1. PREDICTION & MATCHING PIPELINE

The processing engine converts raw legacy material entries into standard, deduplicated canonical master data across six distinct pipeline stages:



2. DETAILED STAGE BREAKDOWN

A. Data Normalization & Cleaning

Standardizes messy legacy entries. Cleans units of measure (e.g., converting "Mtr", "Meters" to standard "M"), formats syntax, and strips unnecessary formatting noise.

B. Engineering DNA 2.0 Extraction

Extracts structured engineering parameters from unstructured material text descriptions into defined parameters: **Function, Material Grade, Dimensions, Rating, and ISO Standards**.

C. Candidate Retrieval & Hybrid Matching

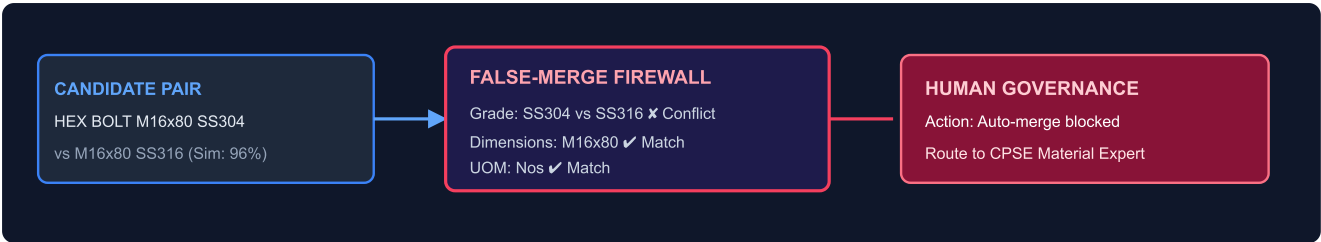
Uses pgvector in PostgreSQL for fast vector indexing (Sentence-BERT embeddings). Top candidates are passed to a pairwise **Cross-Encoder** for deep contextual score calculation.

D. ML Anomaly Detection

A **Random Forest Classifier** evaluates review priorities and outputs confidence anomaly signals to detect potential human review requirements.

3. FALSE-MERGE FIREWALL LOGIC

High text similarity alone can cause catastrophic false-merges in industrial engineering (e.g., merging incompatible metal grades). The False-Merge Firewall applies deterministic safety checks prior to code issuance.



Illustrative Firewall Hard-Block Example

Comparing two highly similar material code descriptions:

Material Record A	Material Record B	Semantic Score	Firewall Decision
HEX BOLT M16 × 80 SS304	HEX BOLT M16 × 80 SS316	96.8% (High)	BLOCKED (Grade Conflict)

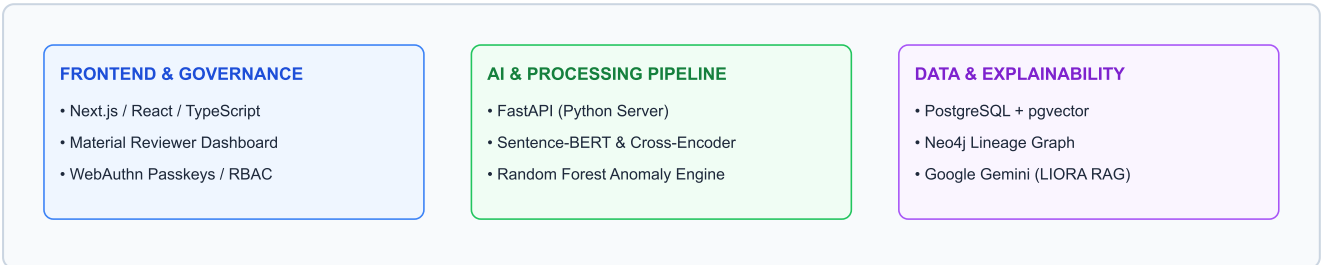
Reason: While dimensions (M16x80) and function (HEX BOLT) match, Material Grade SS304 vs SS316 represents a technical safety conflict. Automatic merging is prevented, routing the record to Human Governance.

4. CURRENT PROTOTYPE PERFORMANCE METRICS

Observed metrics from the connected test environment (evaluated across CPSE labels):



5. PRODUCTION ARCHITECTURE STACK



Layer	Technology Stack	Role in Prediction & Governance
Frontend	Next.js, React, TypeScript	Interactive reviewer dashboard, audit trails, conflict resolution UI
API & Processing	FastAPI, Python Server Routes	Asynchronous execution of ML pipelines and normalization workers
Database & Search	PostgreSQL, pgvector	Relational material master storage with high-dimensional vector index
AI Models	Sentence-BERT, Cross-Encoder, Random Forest	Hybrid semantic retrieval, pairwise contextual scoring, and anomaly detection